

Technologies for E-Commerce

CAN302

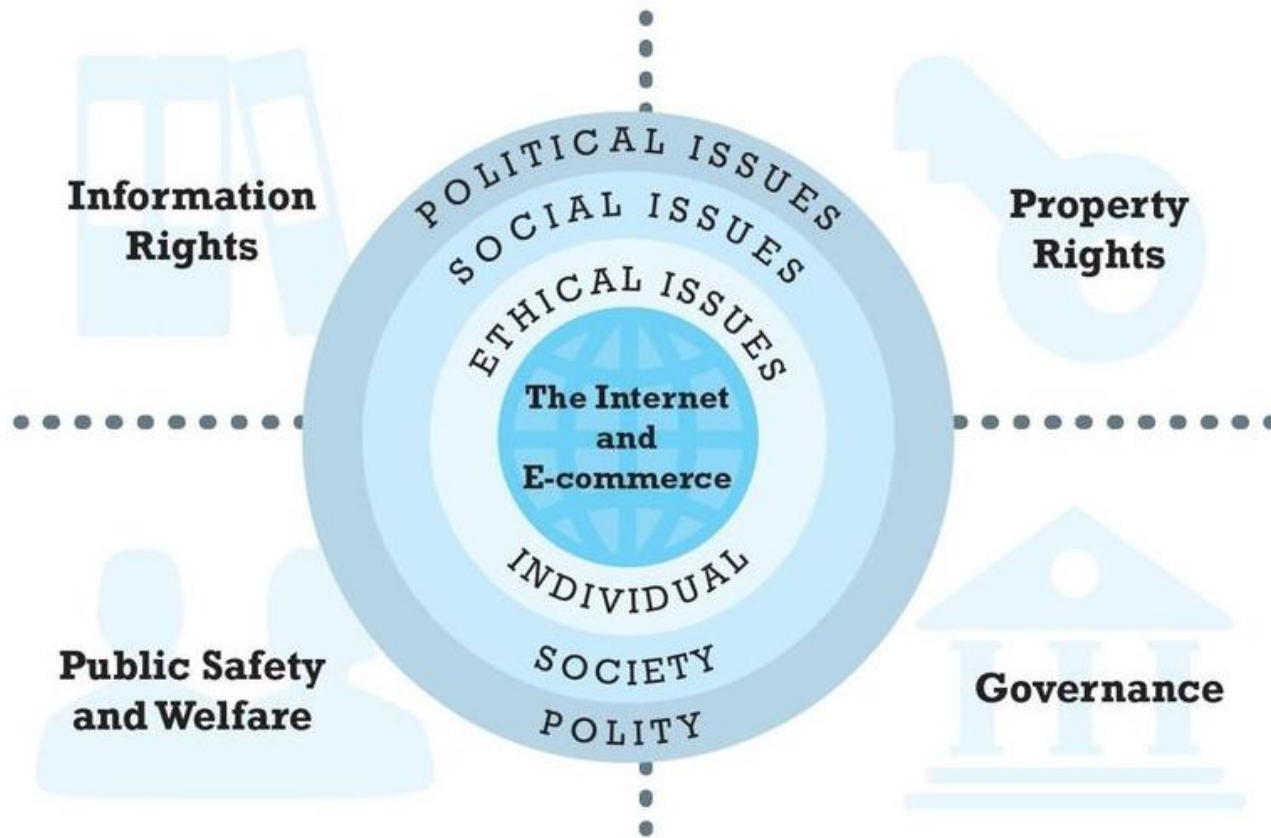
**Department of Communications and Networking
Xi'an Jiaotong-Liverpool University (XJTLU)**

Lecture 9 - Ethical, Social and Digital evidence



Initialed by Google **BUT Abandoned** in 2018!

The Moral Dimensions of an Internet Society



Internet, like other technologies, can:

- Enable new crimes
- Affect environment
- Threaten social values
- Costs and benefits must be carefully considered, especially when there are no clear-cut legal or cultural guidelines

Absolute Privacy is **Not** Possible in any Society



- **Privacy is about the information**
- Privacy is a fundamental human right, but absolute privacy—where no personal data is ever exposed—is impossible.
- At least mum knows her children.
- While the boundary could be very different under different social and technologies background.
- In a village, people know each other very much.
- In a city, people even has very limited information about their neighbors.
- **How about information age?**

Big data even know you better than yourself



- **Internet tools enable the collecting and processing of personal information more powerful than any other time.**
- **Digital Footprints** – Every online action leaves traces, making complete privacy impossible.
- **Surveillance Systems** – Governments and corporations monitor data, limiting true anonymity.
- **Data Sharing** – Apps and services collect & sell user information, eroding confidentiality.
- **Cybersecurity Risks** – Even encrypted data can be breached, exposing personal details.
- **Social Media Exposure** – Voluntary sharing reduces privacy, making secrecy relative.

Big data even know you better than yourself



- It may help to save lives



- It may use to have "**customized**" price!

- **What should the boundary be?**

Intellectual Properties

Type of Intellectual Property



Copyright



Patent



Trademark



Trade Secret



Design



Geographic
Indication

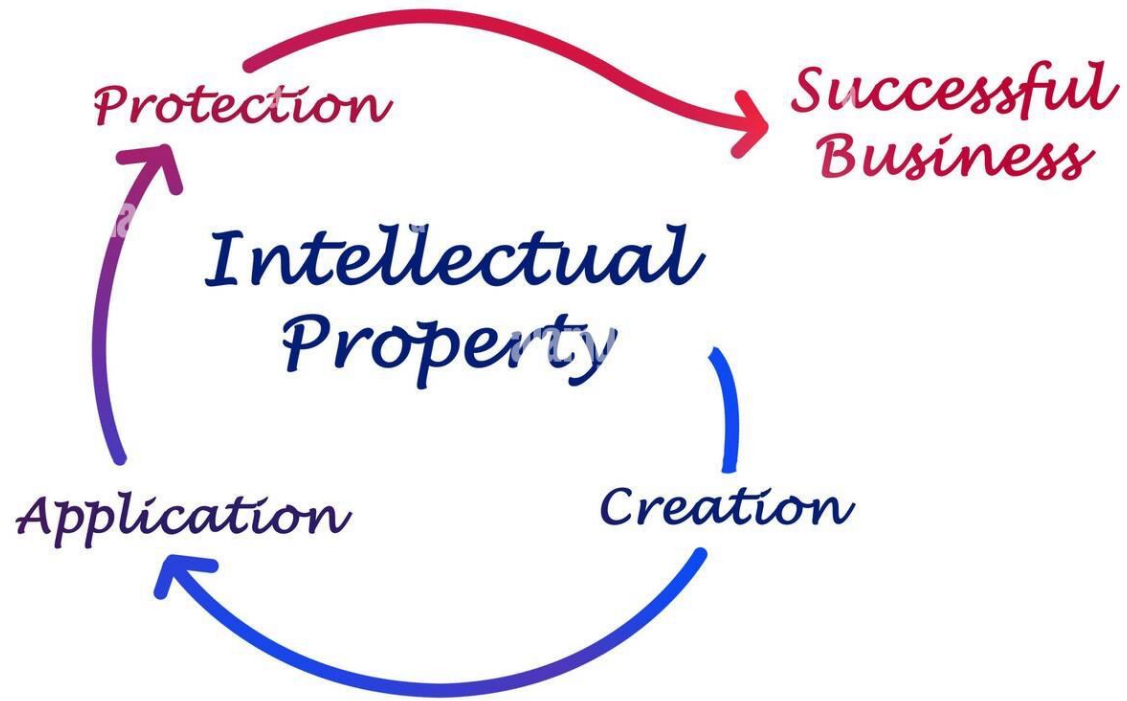


Plant
Variety
Protection

- **Intellectual property (IP)** refers to creations of the mind, such as inventions; literary and artistic works; designs; and symbols, names and images used in commerce.
- They can be classified to different types.
- **Copyright** protects the original forms of expression but not the idea.
- **Patent** grants the owner an exclusive monopoly on the ideas.
- **Trademark** used to identify and distinguish goods and indicate their source.
- Trade secret is secret, has commercial value, and has been protected by its owner.

<https://www.wipo.int/en/web/about-ip/>

Intellectual Properties Protection



- Intellectual property protection drives economic growth **by encouraging innovation** and **securing business value**.
- It offers a competitive edge, allowing companies to protect their market position and maximize the potential of their intellectual assets.
- Technology, healthcare, and entertainment heavily depend on IP to prevent unauthorized use and maintain leadership.
- Protection may have its time limitation. i.e. copyright and patent may expire under certain conditions.

Internet change IP protection a lot



- Internet make the copy of music much easier than previous technologies.

Not realizing the violation

微软雅黑

ABCDEFGHIJKLMNOPQRSTUVWXYZ

0123456789abcdefghijklmnopqrstuvwxyz

微软雅黑

ABCDEFGHIJKLMNOPQRSTUVWXYZ

0123456789abcdefghijklmnopqrstuvwxyz

微软雅黑

ABCDEFGHIJKLMNOPQRSTUVWXYZ

0123456789abcdefghijklmnopqrstuvwxyz

微软雅黑

ABCDEFGHIJKLMNOPQRSTUVWXYZ

0123456789abcdefghijklmnopqrstuvwxyz

微软雅黑

ABCDEFGHIJKLMNOPQRSTUVWXYZ

- These fonts cannot be used for commercial purpose!

无版权字体

宋体、黑体、楷体、隶书、仿宋、篆书，这些字体属于国家公有领域的字体，可随意使用。

免费可商用字体

很多公司都推出了免费可商用的字体，比如思源黑体、站酷酷黑体、阿里巴巴普惠体等。微软雅黑虽然是很多电脑自带的字体，但这并不代表可以免费商用，要使商用记得交版权费！

自己设计的字体

可以自己设计花字，设计出切合品牌形象的文字商标。但使用不规范汉字，如自造字、缺笔画或笔画错误等，很可能就会被驳回。

duāng
成

福

嘉逸达

How about this case



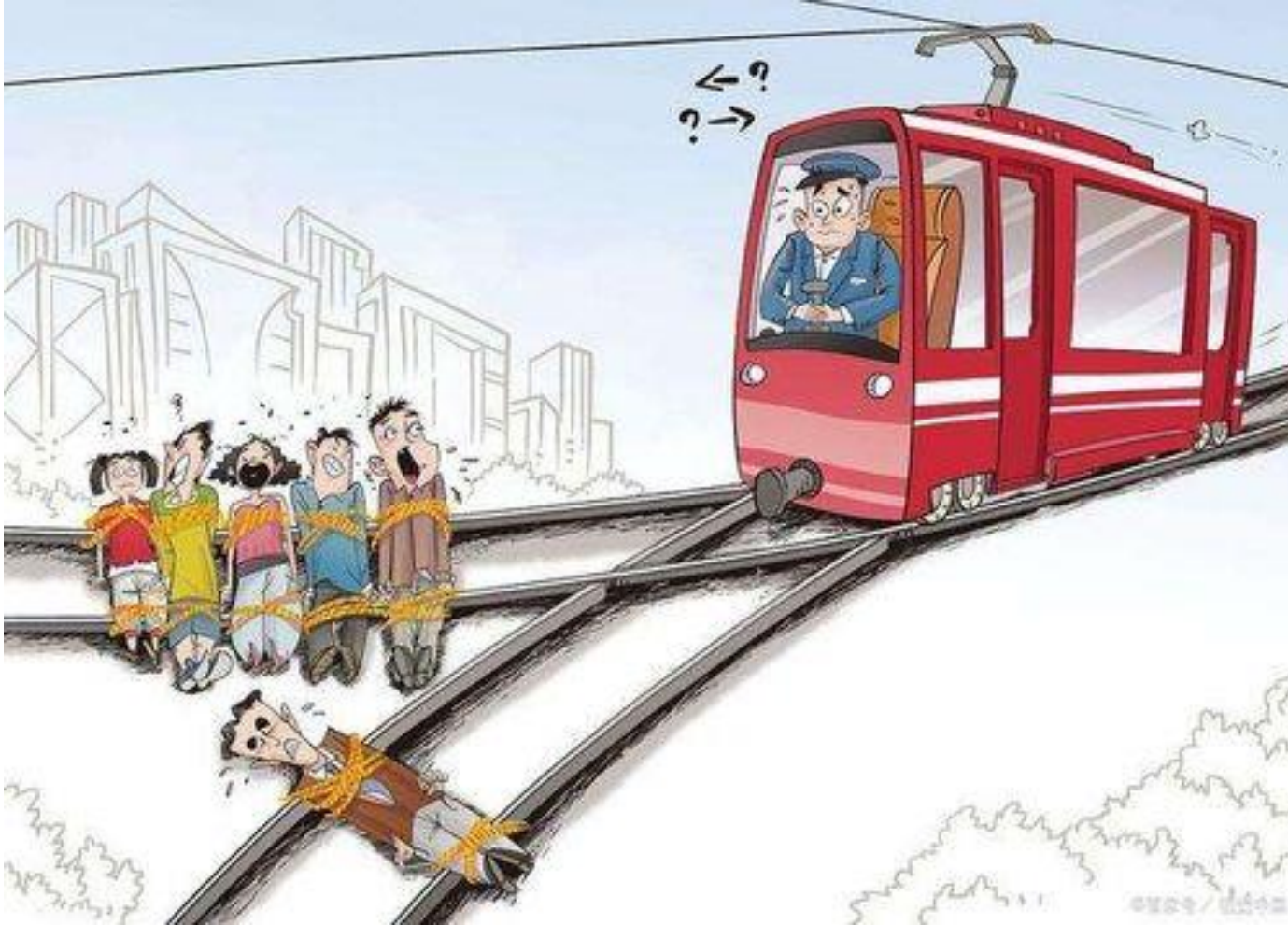
- VCG even claim the flag is their copyright!

No absolute IP protection



- Intellectual property protection is not absolute.
- **What should be boundary be?**

Trolley Problem



- **Jeremy Bentham** 1748 – 1832
- **Utilitarianism** is a family of normative ethical theories that prescribe actions that maximize happiness and well-being for all affected individuals.
- **Immanuel Kant** 1724 – 1804
Deontological ethics or deontology is the normative ethical position that judges the morality of an action based on the action's adherence to a rule or rules.

Trolley solutions



- Social contract theory
- Virtue theory
- 无为
- 齐身修家治国平天下

- No simple answer to the complex questions

<https://www.bilibili.com/video/BV13a411b7Ry>

What is the law?



- **Law** is a system of rules created and **enforced** through social or governmental institutions to regulate behavior, with its precise definition a matter of longstanding debate.
- **Law** has been variously described as a science and the art of **JUSTICE**.

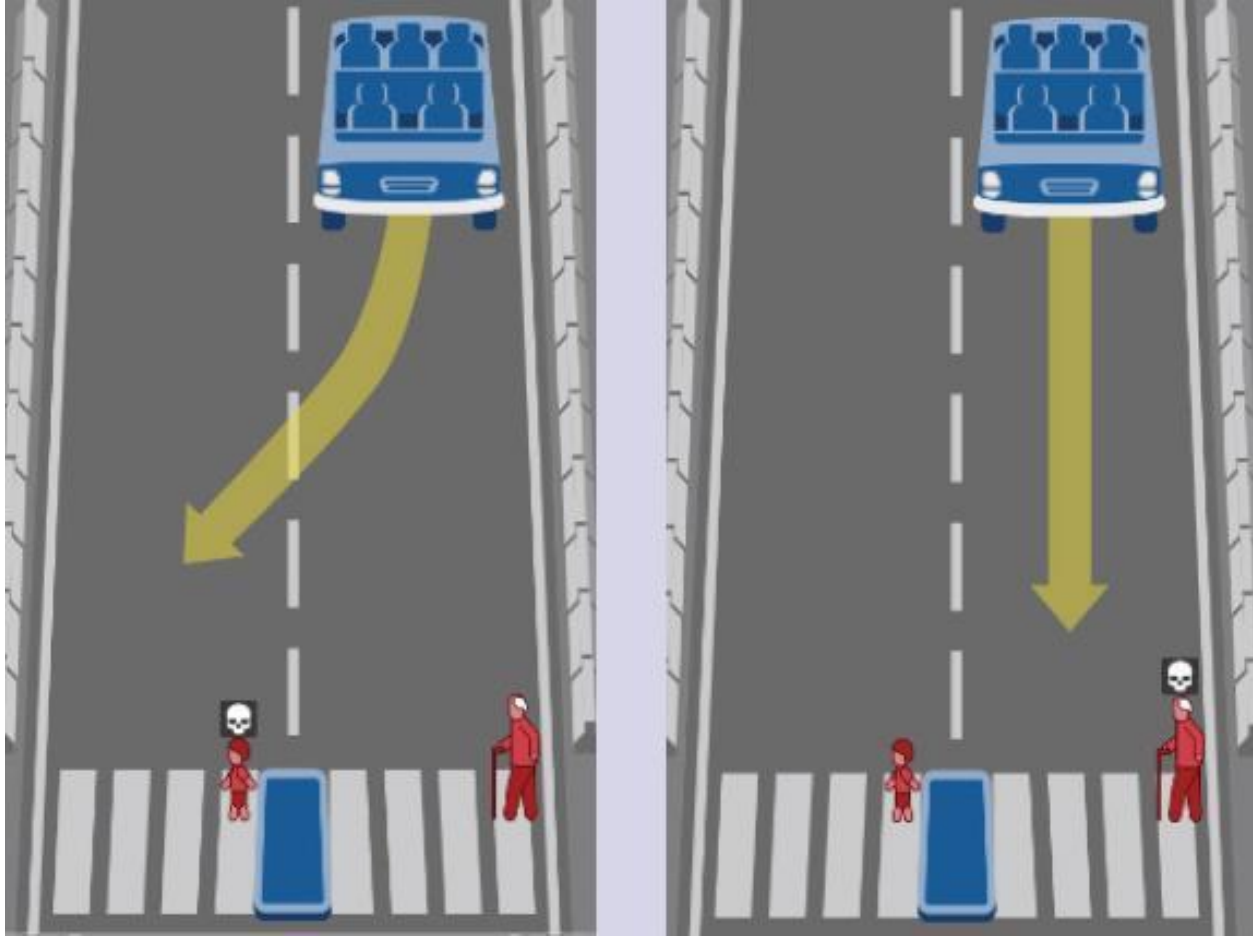
Law and ethic



金剛怒目所以降伏四魔
菩薩低眉所以慈悲六道
沒有霹雳手段
難显菩薩心肠

- Both ethics and laws are found in virtually all spheres of society. They govern actions of individuals around the world on a daily basis.
- Laws **enforce** the behaviors we are expected to follow.
- While ethics **suggest** what we ought to follow.

Legal and ethic



- Human has no pre-definition for this condition.
- AI algorithm must make the choice!
- Law is always late than the scenarios.

Do not be evil



- Do unto others as you would have them do unto you. – A Gentleman/woman!
- Do you think I would break the law? Absolutely not! I'm very lawful! –

<https://baijiahao.baidu.com/s?id=1728262469984324020>

Do not be evil



- They are all!

Do not be evil

工业和信息化部通报存在问题的应用软件名单

序号	应用名称	应用开发者	应用来源	应用版本	所涉问题
1	腾讯新闻	深圳市腾讯计算机系统有限公司	应用宝	6.6.51	欺骗诱导强迫用户 应用分发平台上的 APP 信息明示不到位
2	全民 K 歌极速版	腾讯音乐娱乐科技(深圳)有限公司	应用宝	7.7.28.278	强制用户使用定向推送功能 欺骗诱导强迫用户
3	QQ 音乐	深圳市腾讯计算机系统有限公司	应用宝	10.18.5.9	超范围收集个人信息
4	小红书	行吟信息科技(上海)有限公司	应用宝	7.13.0	超范围收集个人信息
5	探探	探探文化发展(北京)有限公司	应用宝	4.8.8.2	超范围收集个人信息
6	作业帮	小鹅出海教育科技(北京)有限公司	应用宝	13.25.2	超范围收集个人信息
7	看看新闻	看东方(上海)传媒有限公司	应用宝	6.1.7	超范围收集个人信息 APP 强制、频繁、过度索取权限
8	我的日程表	有鲤科技(厦门)有限公司	应用宝	1.2.4	超范围收集个人信息
9	亚朵	上海亚朵商业管理(集团)股份有限公司	360 手机助手	3.13.0	违规收集个人信息 超范围收集个人信息 违规使用个人信息 强制用户使用定向推送功能 欺骗诱导强迫用户
10	影视大全	襄阳品山信息科技有限公司	360 手机助手	4.2.5	超范围收集个人信息 欺骗诱导强迫用户 应用分发平台上的 APP 信息明示不到位
11	宝宝树孕育	宝宝树(北京)信息技术有限公司	vivo 应用商店	8.60.0	超范围收集个人信息
12	妈妈网孕育	广州盛成妈妈网络科技股份有限公司	vivo 应用商店	11.10.3	超范围收集个人信息
13	动动	北京百索科技有限公司	vivo 应用商店	8.7.1.1	超范围收集个人信息
14	嗨呀星球	武汉佐趣科技有限公司	vivo 应用商店	2.3.6	APP 强制、频繁、过度索取权限
15	豆豆免费小说	北京友和卓谊信息技术有限公司	vivo 应用商店	5.6.0	APP 强制、频繁、过度索取权限 欺骗诱导强迫用户
16	微商水印 Pro	北京助梦工场科技有限公司	vivo 应用商店	5.2.88	APP 强制、频繁、过度索取权限
17	班级优化大师	广州视睿电子科技有限公司	vivo 应用商店	3.0.33.1	APP 强制、频繁、过度索取权限
18	北京搜狐新媒体信息技术有				APP 强制、频繁、过度索取权限



Do no be evil



https://m.thepaper.cn/baijiahao_16598227

Battle between Giants



- When the growth is ending, it always the time for a WAR!
- Round I : PC age

<https://baike.baidu.com/item/%E8%85%BE%E8%AE%AF360%E4%B9%8B%E4%BA%89/7181338>

Battle between Giants



- When the growth is ending, it always the time for a WAR!
- Round II : mobile age

Some laws in Europe and UK

- 1. Getting Europe ready for the digital age: updates to the proposed Digital Services Act
- 2. Getting Europe ready for the digital age: updates to the proposed Digital Market Act
- 3. UK's New Digital Task Force targets Big Tech
- 4. Upcoming EU legislation will address AI, robots, and machine learning – and liability issues
- 5. Geo-blocking continues to be a key area of development and enforcement in the EU
- 6. EU regulation of Online Intermediaries begin to takes shape
- 7. Digital consumer law reform in the UK
- 8. European Electronic Communications Code: Delayed implementation extends uncertainty for providers
- 9. UK & German Governments Respond on Online Harms
- 10. Digital Copyright: New (case) law

<https://www.mofo.com/resources/insights/210414-key-digital-regulation.html>

Some laws in China

- **法律**

- 网络安全法
- 数据安全法
- 个人信息保护法
- 电子商务法
- 电子签名法
- 全国人大常委会关于加强网络信息保护的決定
- 全国人大常委会关于维护互联网安全的決定

- **法规**

- 互联网信息服务管理办法
- 互联网上网服务营业场所管理条例
- 信息网络传播权保护条例
- 计算机信息网络国际联网安全保护管理办法
- 计算机信息系统安全保护条例
- 计算机软件保护条例
- 电信条例
- 网络数据安全管理条例
- 关键信息基础设施安全保护条例

- And many more……

<https://zhuanlan.zhihu.com/p/465031437>

The golden rule

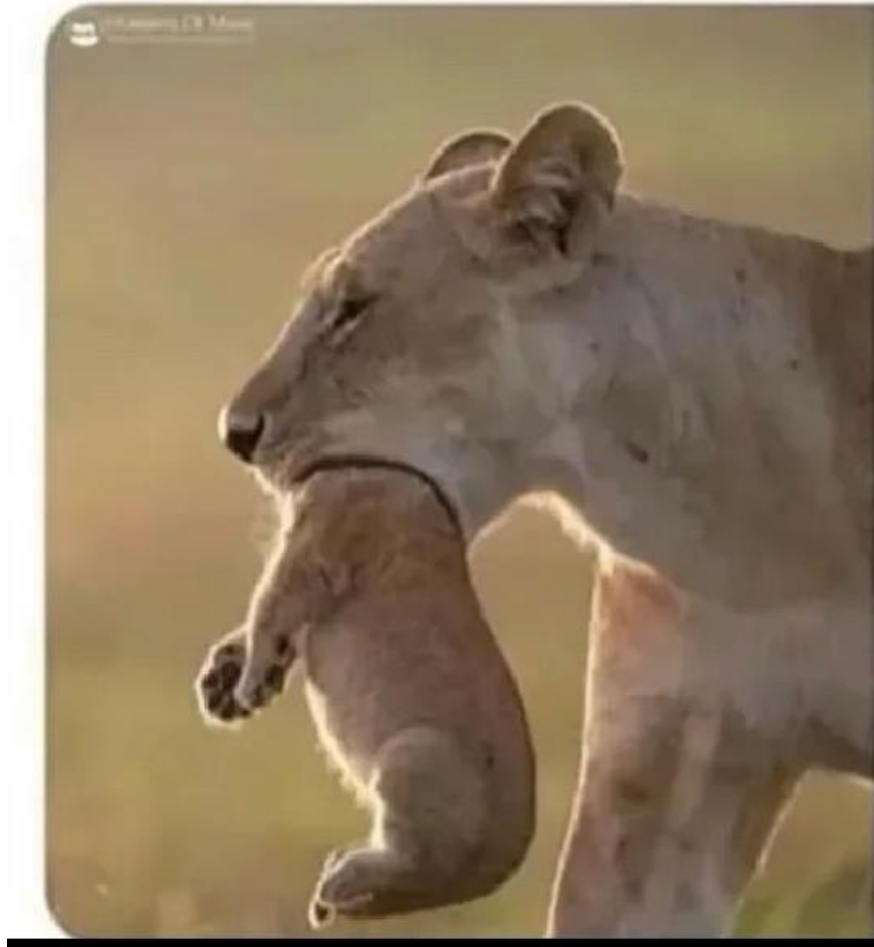
“Don’t Be Evil”

The importance of evidences



- The record is only a part of the truth!
- Courts can only "recovery" the truth based on the records

The importance of evidences



- Black or white?
- It depends on the records.

The importance of evidences



- Truth in the court may not be the truth in the life
- How to let the court believe your evidences?

https://m.sohu.com/a/334229522_100114073

The importance of evidences



Digital evidence

Sources of Electronic Evidence

Where is the Evidence?

Physical Sources

Computers, Servers, Phones, NAS, Flash Drives, Tape Backups, Phones, CD/DVDs, Flash Drives, External Hard Drives, Smart Watches, etc.

BYOD

Electronic Sources

Logs, Emails, File Sharing

External Sources

Websites, Apps, Cloud, Social Media, etc

- Digital evidence is **fragile** and
- can be altered, damaged, or destroyed by improper handling or examination.

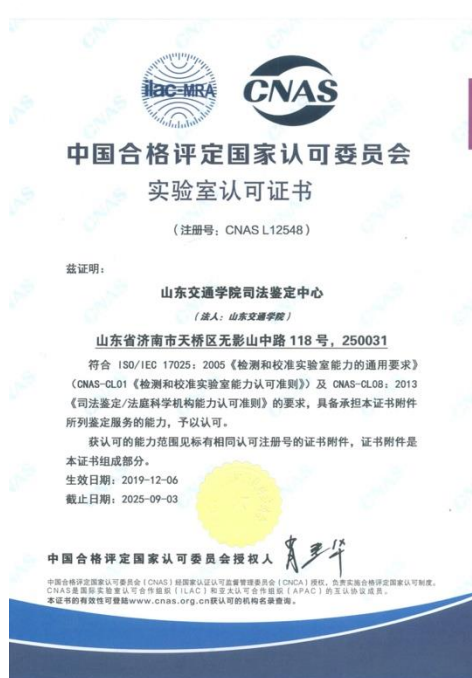


Collecting data

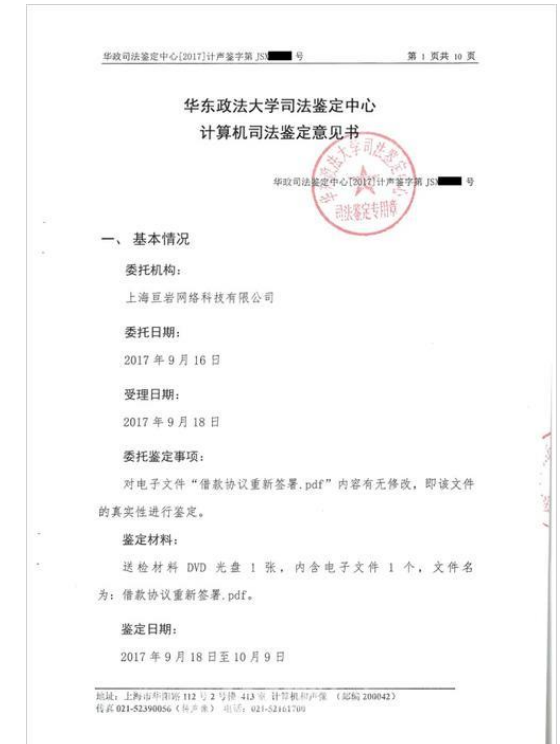


- Only authorized department can collect data in this way

Judicial Expertise

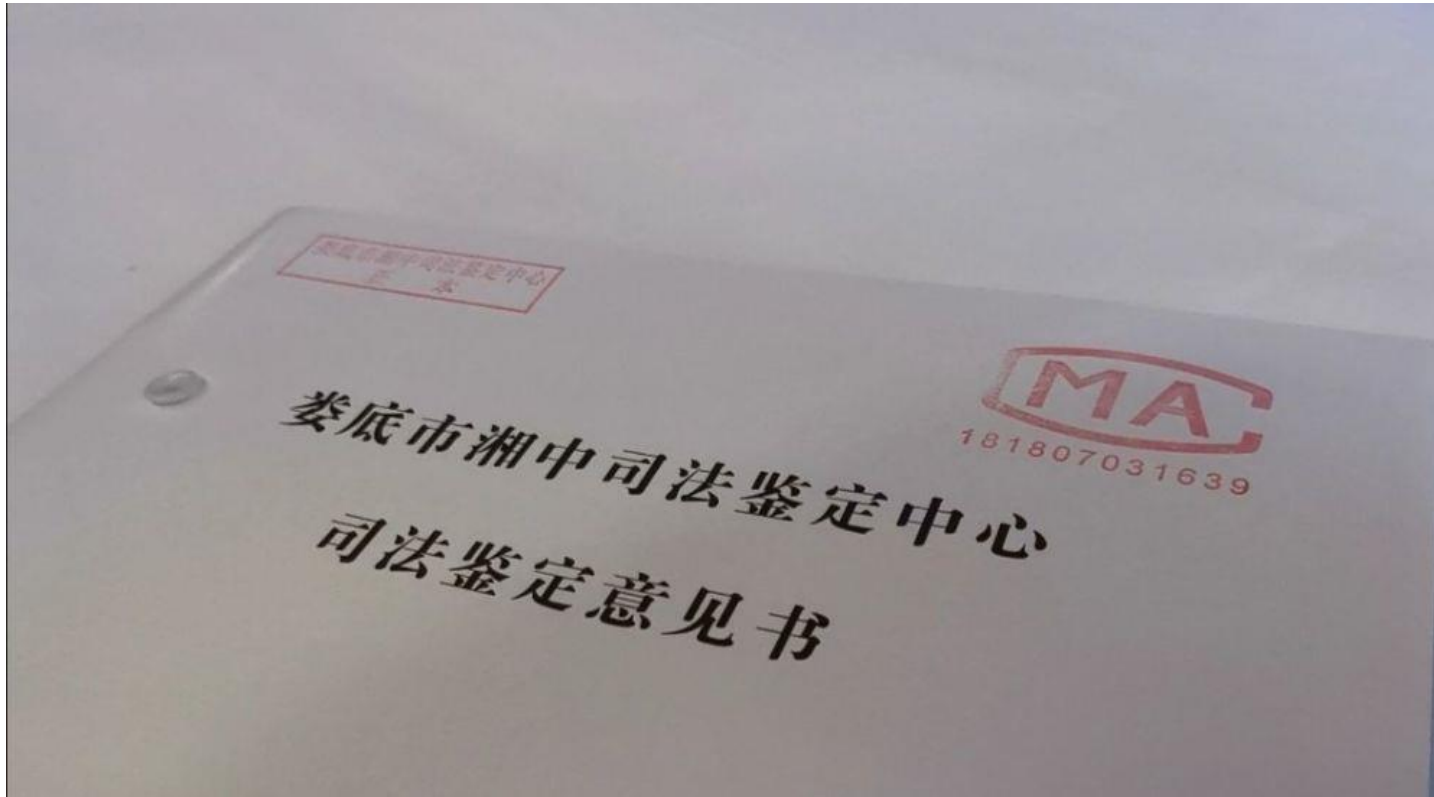


中华人民共和国司法部监制



- Judicial Expertise center is a lab with the judicial permission
- Expert Report by Judicial Expertise can **increase** the creditable
- **Cannot** work for the **disappeared** evidence

Judicial Expertise



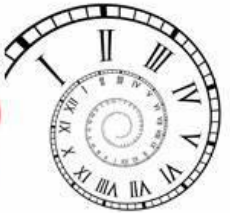
<https://www.bilibili.com/video/BV1uY4y1F7Np>



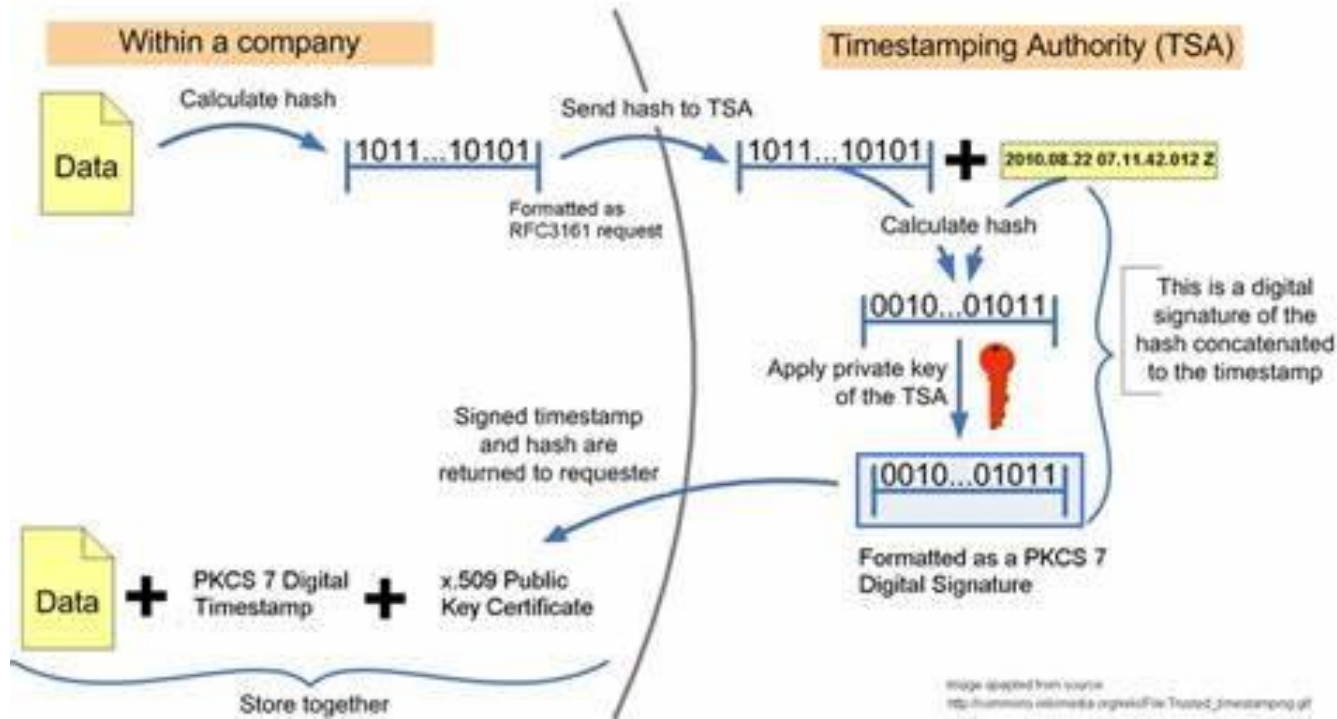
- Exam the DNA of parrots by pictures...

Preservation of evidence

time
stamp

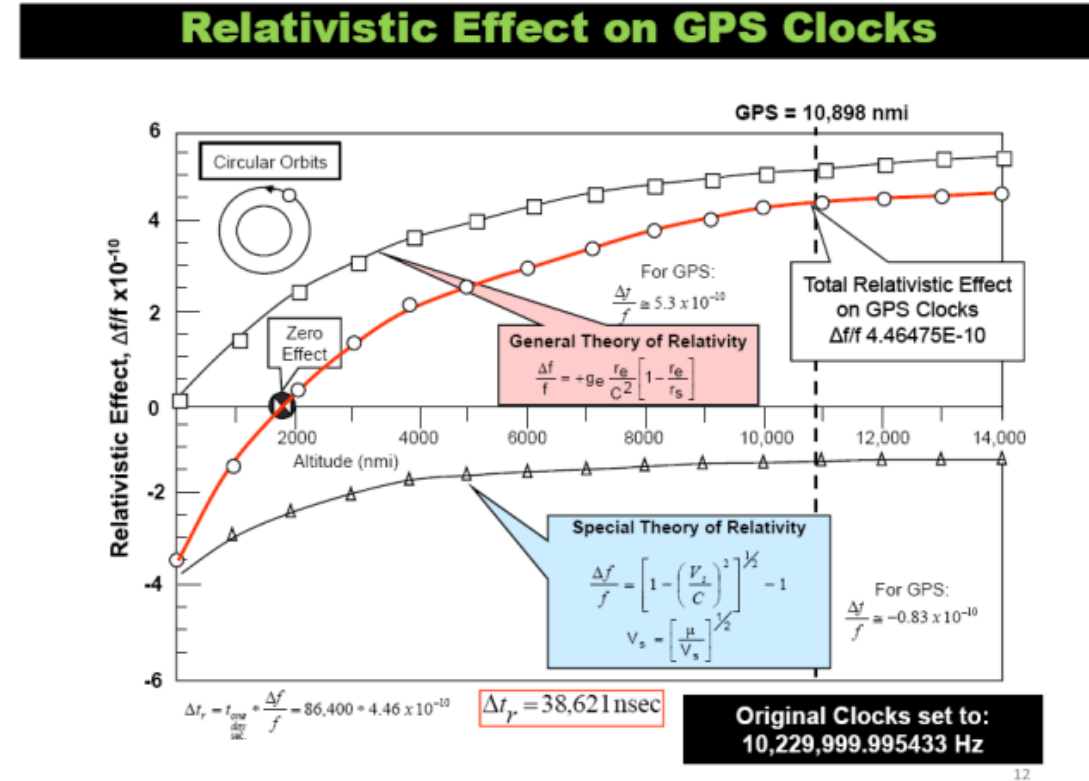
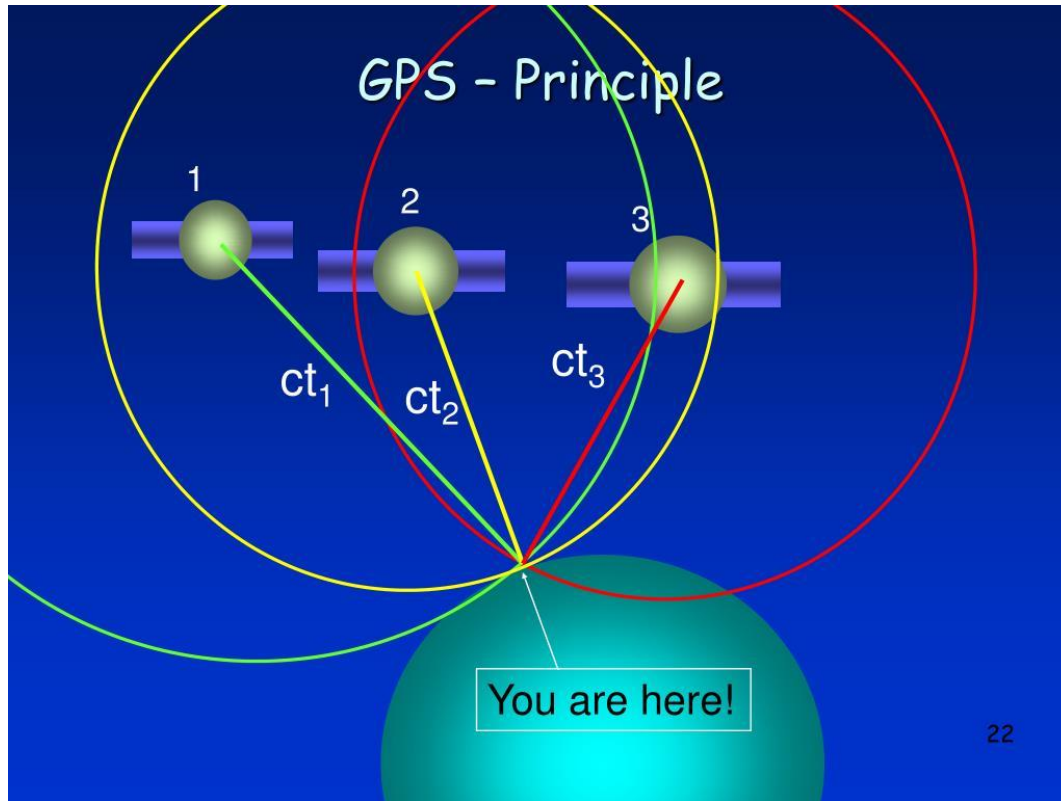


capturing this moment in time



- TSA should be the authorized organization

Get time from navigation satellites



- The clock need to be synchronized with the grand station.

Certificates for TSA server

 中国科学院国家授时中心

检测证书
证书编号: NTSC SFJ1101-J2

送检单位: 时间用户系统研究室
设备名称: 时间戳服务器
型号规格: SFJ1101
制造厂商: 北京安信天行科技有限公司

测试人: 朱青
验证人: 张涛
检验员: 检01

(质检专用章)

检测日期: 2012 年 12 月 27 日
检测单位: 中国科学院国家授时中心
地址: 西安市临潼区书院东路 3 号 邮编: 710600
电话: 029-83890287
传真: 029-83890326

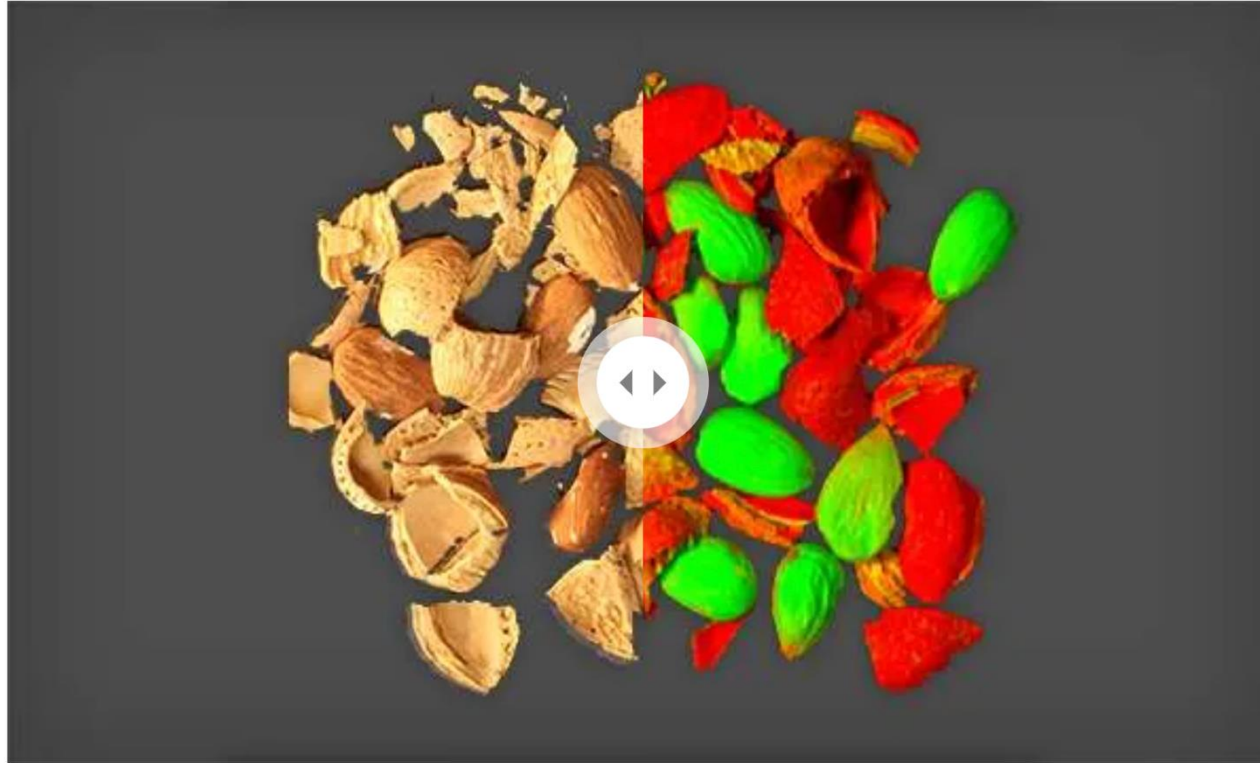
商用密码产品型号证书
(证书编号: SXH2011123 号)

单位名称: 北京安信天行科技有限公司
批准型号: SFJ1101
产品名称: 时间戳服务器

国家密码管理局
2011 年 7 月 8 日

- TSA servers' certificate
- The **"official time"** for the birth of evidence

Tampered or not?



■ Shell ■ Nut (Almond)

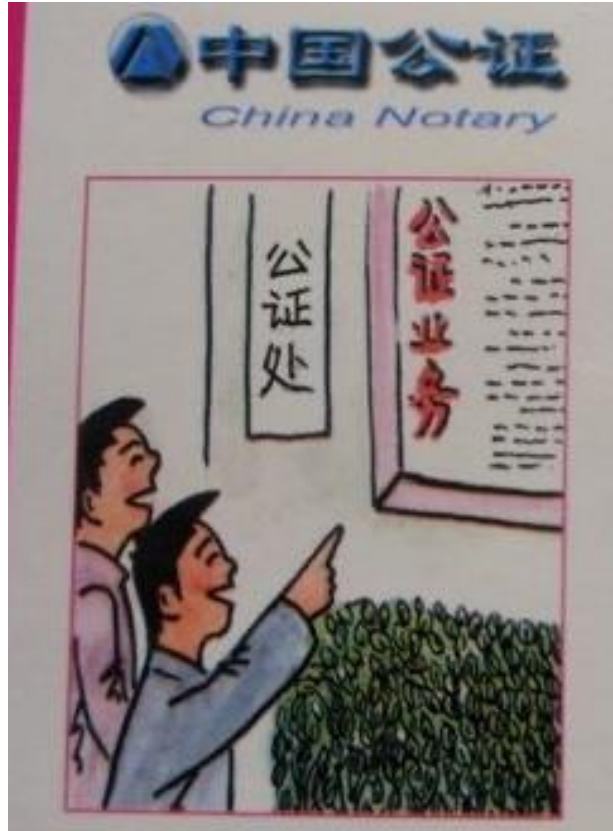
- Does the data is tamped or not before TSA?
- Nobody can tell it easily.

Preservation of evidence



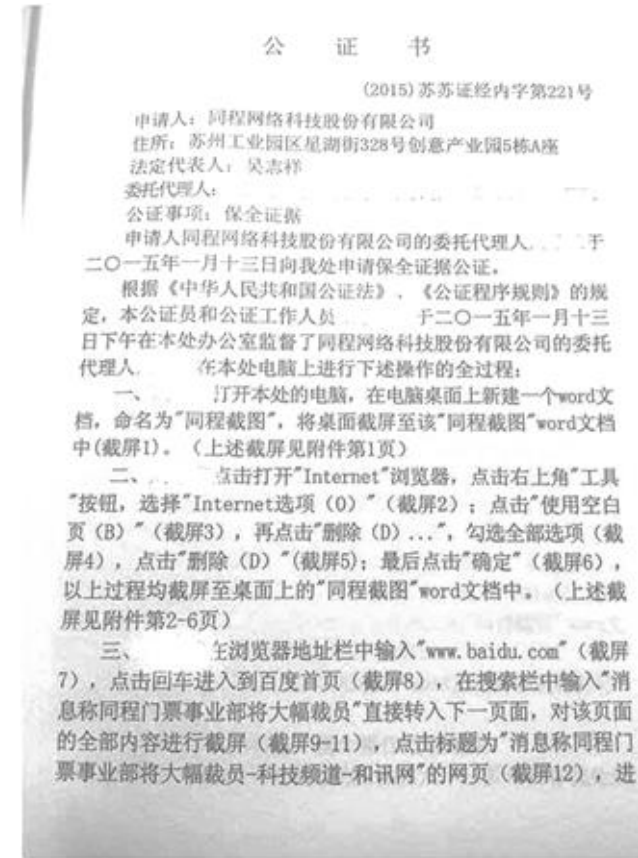
- Let witnesses to prove the data is true
- Should we trust on her words?

The notarial institute



- Notarial officers are special witnesses
- Notarial report have a strong creditable by the law

Notarial preservation of evidence



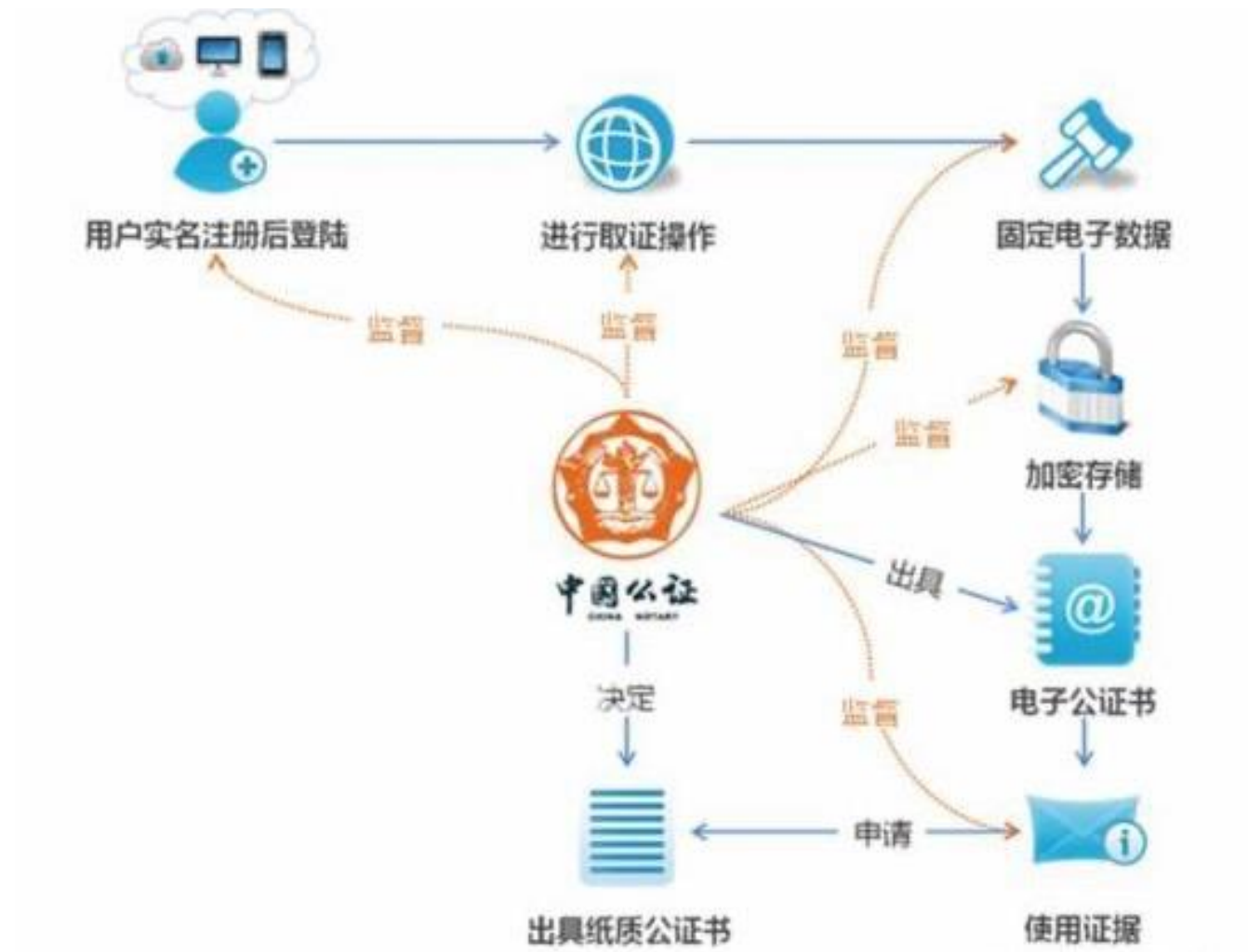
- Full process under surveillance
- Using the computer and network provided by the notarial institute
- Reduce the unsure factors

Notarial preservation of evidence



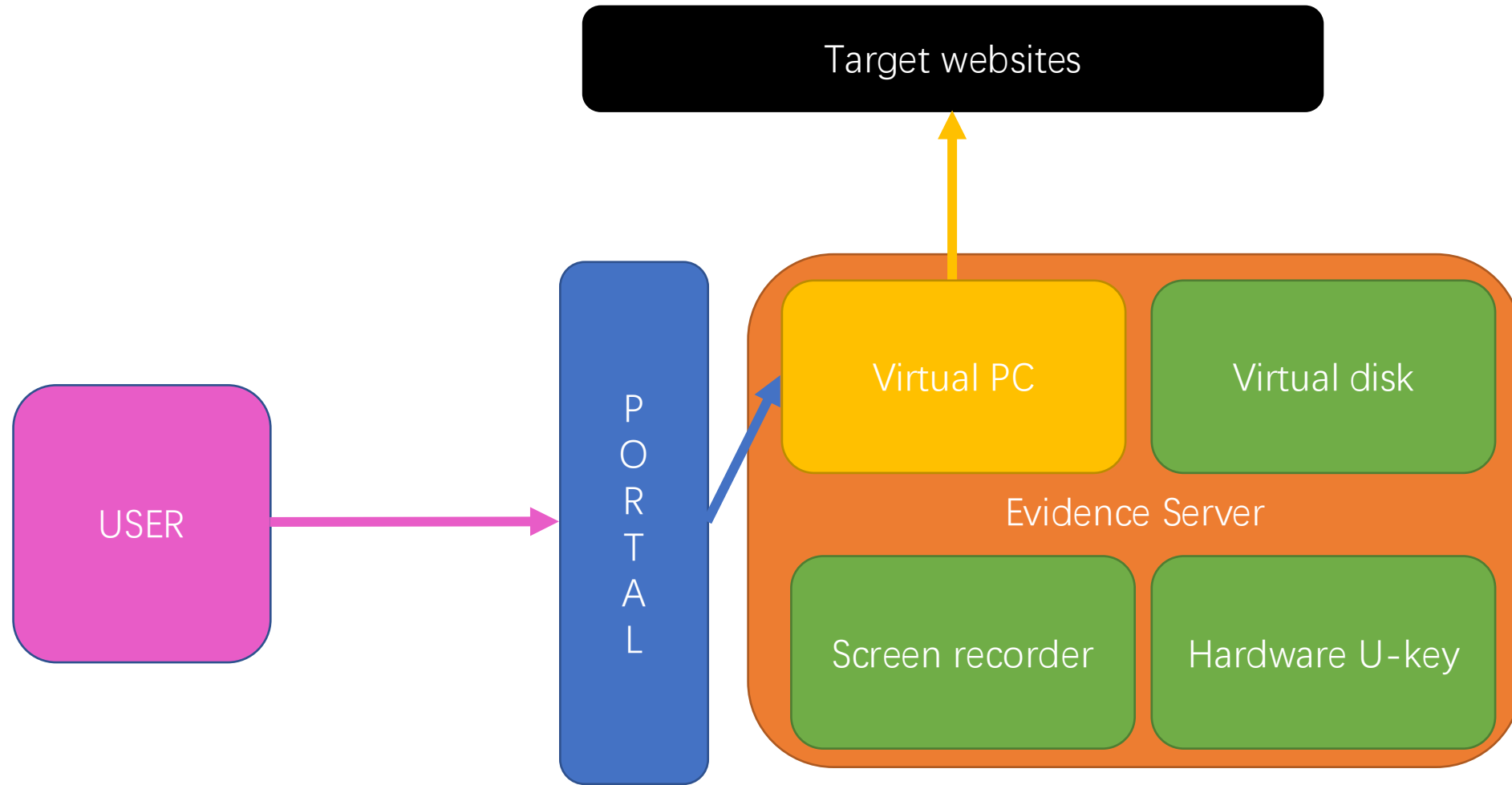
- Manually service cannot satisfy the 7x24 hours demands
- For example, living broadcast of European football games

Preservation of evidence



- Special cloud for preservation purpose

Preservation of evidence



- The notarial cloud for preservation purpose

Preservation of evidence



- Blockchain is a hot technology.
- Organizations in the same chain can easily share the evidences.
- While, if one organization hold the chain, the credit cannot beyond the credit of that organization.